SIMATS ENGINEERING
ASSESSMENT TOOL 1

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO2	Assessment	Analytical Problem Solving
Weightage		Total Marks	

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Section/Batch	AIDS-2023	Date of Submission	
Faculty Incharge	DR.R S Selvan	Assessment	Individual Submission

TOTAL MARKS	20	OBTAINED MARKS	

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

17/20

Code of Conduct:

I V. Sree Ruthin Reddy (192324112) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. R

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 **subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Subnetting 192·168·20·0/24

Given, original network = 192·168·20·0/24

Lab A - /26

Subnet mask

/26 = 255·255·255·192

Block size = 256-192
= 64.

So each /26 subnet contains 64 addresses

starting from 192·168·20·0

Network address = 192·168·20·0

~~Neetw~~ first Host = 192·168·20·1

Last Host = 192·168·20·62

Brodcast Address = 192·168·20·63

Usable hosts :

$2^{32-26} - 2 = 2^{6} - 2 = 64 - 2$
$= 62$,,

Lab B - /27

The next unused address after Lab A is

192·168·20·64

Subnet mask /27 = 255·255·255·226.

Block size : 256 - 224 = 32

Subnetting of two systems.

Given :

System 1 = 192.168.1.130

System 2 = 192.168.1.190

Subnet mask = 255.255.255.196.

This is a /26 mask.

Block size : 256 - 192 = 64.

Therefore, Subnet ranges are :

0-63, 64-127, 128-191, 192-255

Both 130 & 190 fall in the 128-191 subnet

System 1 : 192.168.1.130

Subnet range :

192.168.1.128 - 192.168.1.191

Therefore

Network Address = 192.~~168~~.1.128

First Host = 192.168.1.129

Last Host = 192.168.1.190

Broadcast Address = 192.168.1.191

Since :

$129 \leq 130 \leq 190$

192.168.1.130 is a Valid host IP.

Therefore,

Network address : 192.168.20.64

First host = 192.168.20.65

Last host = 192.168.20.94

Broadcast Address = 192.168.20.95

usable hosts : $2^{32-27} - 2 = 2^5 - 2$

$= 32 - 2$

$= 30$

Lab C - /28.

The next unused address is 192.168.20.96

Subnet mask:

128 = 255.255.255.240

Brock size:

256 - 240 = 16

Therefore

Network Address = 192.168.20.96

First Host = 192.168.20.97

Last Host = 192.168.20.110

Broadcast address = 192.168.20.111.

usable hosts :

$2^{32-28} - 2 = 2^8 - 2$

$= 16 - 2 = 14.$

System 2: 192.168.1.190.

190 also belongs to the same 128-191 subnet

Therefore:

Network Address = 192.168.1.128

First Host = 192.168.1.129

Last Host = 192.168.1.190.

Broadcast Address = 192.168.1.191

Since 190 is the last valid host address,

192.168.1.190 is a valid host IP.

usable hosts : $2^{32-26} - 2 = 2^6 - 2$

$= 62$

∴ usable hosts per subnet = 62

∴ Both systems belong to the same Subnet

3. Subnetting 172.16.0.0/16

Given:

HR → 125

Sales → 126

IT → 127

Admin → 128.

HR → 128.

Subnet mask = 255.255.255.128.

Total addresses : $2^{32-25} = 2^7$

$= 128$

usable :

128 - 2 = 126.

Therefore

Network = 172.16.0.0

Host range = 172.16.0.1 — 172.16.0.126

Broadcast = 172.16.0.127

Sales - 126

Nextwork = 172.16.0.128

Total addresses = 2^{32-26}

= 64

Usable = 64 - 2

= 62

Therefore

Network = 172.16.0.128.

Host range = 172.16.0.129 — 172.16.0.190

Broadcast = 172.16.0.191

IT - 127

Next available address = 172.16.0.192

Total addresses : 2^{32-27} = 32

usable : 32 - 2 = 30,,

Therefore :

Network = 172.16.0.192

Host range = 172.16.0.193 - 172.16.0.222

Broadcast = 172.16.0.223.

Admin - /28

Next available address = 172.16.0.224

Total addresses : 2^{32-28}

= 16

usable = 16 - 2 = 14

Therefore :

Network = 172.16.0.224.

Host range = 172.16.0.225 — 172.16.0.238.

Broadcast = 172.16.0.239

The last allocated address is 172.16.0.239.

Therefore, the unused addresses begin from:

172.16.0.240

The compeete /16 block ends at : 172.16.255.255

Subnetting 172.16.10.0/24 using /27

Original network = 172.16.10.0/24

New subnet = /27

No. of Subnets

Original bits = 24

New bits = 27

Borrowed bits : 27 - 24 = 3.

No. of subnets : $2^3 = 8$ Subnets

No. of hosts

Host bits : $32-27$
$= 5$

usable hosts : $2^5 - 2$
$= 32 - 2 = 30$

Find subnet containing 172.16.10.78

/27 subnet mask : 255.255.255.244.

Block size $= 256 - 224$
$= 32$

Subnets ranges are :

0-31, 32-63, 64-95, 96-127, 128-159, 160-191, 192-223, 224-255.

IPv6 Address compression

Given IPv6 address,

2001:0d68:0000:0000:0000::ff00:0042

Remove leading zeros.

IPv6 allows leading zeros in each group to removed.

So

2001:db8:0:0:0:ff:00:42:8329.

Repeace consecutive zero groups.

The three consecutive 0 groups can be represented by ::

Therefore the shortest form is

2001:db8::ff00:42:8329.

Expand it back.

Compressed :

2001 : 0db8 : 0000 : 0000 : 0000

Therefore, 64/16 = 4 groups.

First four groups :

2001 : 0db8 : 0000 : 0000.

So, the newest prefix can be written as :

2001:db8:0:0::/64.

Unlike traditional IPv4 subnet calculation, we normally dont subtract 2 for network & broadcast addresses because IPv6 doesnot use a broadcast addresses.